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The study of the electron transport characteristics of bilayer blue phosphorus with different stacking by first principles

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Abstract

We study the structural, electronic and transport property of bilayer blue phosphorus (BBP) by using the first-principles. Our results show that the band gap can be adjusted by different stacking structures of the BBP. We simulate the functional device based on AA-, AB- and AC-stacking BBP and the transport characteristics of the current-voltage curve with nonlinear competitive behavior are investigated. Of the three devices, AA stacking BBP has the highest conductivity. Under special bias, the currents of AB- and AC-stacking devices produce interesting competitive behavior. The transport characteristics behaviors of the BBP can be explained by the band structure, transport spectrum and molecular projected self-consistent Hamiltonian. We can control the change of current by adjusting the different contact modes of the BBP. The BBP with interesting electronic and transport properties are expected to have potential applications in nanoelectronics.

Keywords: first principles, blue phosphorus, electron transport, transport spectrum, electronic orbital

 Supplementary material for this article is available [online](#)

(Some figures may appear in colour only in the online journal)

Introduction

Two-dimensional (2D) materials have attracted huge interest and led to a prosperous development in both fundamental and applied studies in various fields [1–5]. Among various 2D semiconductors, due to the unique properties of phosphorene [3–5], such as a direct band gap [5, 6] and high carrier mobility ($\sim 1000 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) [3], this new 2D semiconductor

has high potential to be the core material of next generation nanoelectronic devices. Black phosphorus is the most stable phosphorus and layered structure of an elemental solid as graphene. Individual layers of black phosphorus have a non-planar honeycomb structure.

As the allotrope of black phosphorus, Zhu *et al* studied its stability by first principle methods and found that it can be as stable as black phosphorene [7]. The new 2D material,

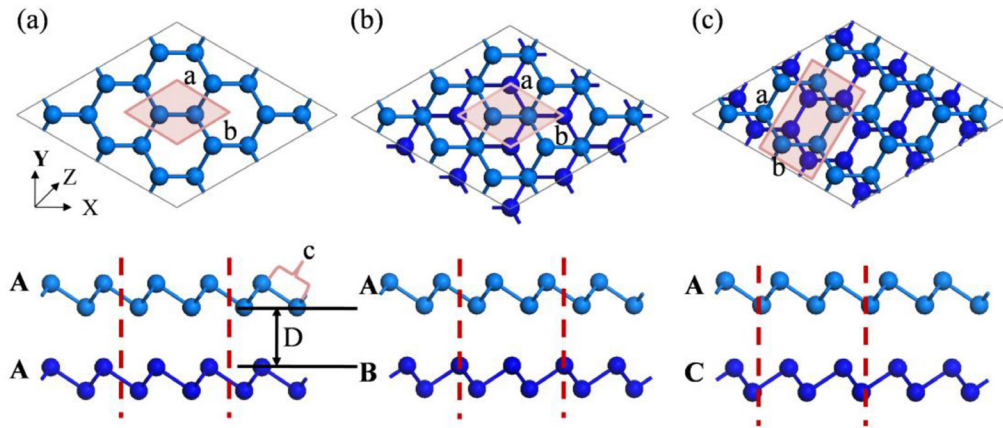


Figure 1. Top and side views of three kinds: (a) AA-, (b) AB- and (c) AC-stackings constructions bilayer blue phosphorene.

Table 1. Equilibrium structure parameters of the monolayer and BBP.

Type	a (Å)	b (Å)	c (Å)	D (Å)	E_b (eV)	E_g (eV)	CBM (eV)	VBM (eV)
Monolayer	3.32	3.32	2.2912		0	1.890	0.938	0.955
AA	3.267	3.267	2.2711	3.214	0.230	0.987	0.442	0.545
AB	3.267	3.267	2.2705	3.248	0.231	1.100	0.524	0.576
AC	3.267	5.659	2.2709	3.262	0.210	1.055	0.509	0.546

known as blue phosphorous (BlueP), has a cell of two sublattices that resemble a honeycomb lattice but are not on the same horizontal plane (see figure 1(a)), forming a periodically buckled structure. Unlike zero band gap graphene, BlueP possesses 2.0 eV energy gap, which is considerably higher than that of black phosphorus [7, 8]. Recently, Xu and Zhang have successfully grown a single layer of blue phosphorus on the Au(111) membrane by molecular beam epitaxy (MBE) [9, 10]. Unlike other 2D materials, such as graphene [11] and transition metal dichalcogenides [12], the phosphorus atoms bond with three neighboring phosphorus atoms to form a special, buckled structure. Based on the first principles calculation, a method of modulating the blue band gap by adjusting the number of blue phosphorus layers is proposed [13]. The energy gap is increased when the layers number is decreased. Another way to modulate the physical properties of BlueP is to change the two-layer stack. Two layers of BlueP stacked AA, AB, or AC also have a smaller band gap than a single layer. [19].

In our work, we focus on the electrical property of pure bilayer BlueP by density functional theory (DFT) calculation [14, 15]. We calculate the band structures of different stacked bilayer BlueP and simulate the sandwich devices based on bilayer BlueP. The transport properties are studied through DFT combined with a nonequilibrium Green's function (NEGF) [16–18], where atomic relaxations of the whole system are taken into account. We will study AA-, AB- and AC-stacking BlueP.

Theory and methods

The calculations are based on the Virtual Atomistix Toolkit (ATK) [16–18]. The DFT calculations of bilayer BlueP are

performed with the Perdew–Burke–Ernzerhof (PBE) and generalized gradient approximation (GGA).

In order to expend the electron density, a double- ζ polarized basis set is used. We sample the $12 \times 12 \times 6$ K-points in the Brillouin zone (BZ) of the bilayer BlueP bulks. The maximum force is $0.05 \text{ eV } \text{\AA}^{-1}$ and the density mesh cut-off is 150 Rydberg during geometry optimizations. We also used local density approximation (LDA) calculation, and got a similar result. We calculate the electronic property of different stacking bilayer blue phosphorus (BBP) as summarized in figure 1. In order to avoid spurious interactions caused by the periodic boundary conditions, a vacuum region with a thickness of $15 \text{ \AA}$ along the z direction is introduced. The side view of the BBP can be seen in figure 1. In this work, we have optimized both the super cell and the lattice to obtain the most stable atomic positions.

Results and discussion

All structural parameters are listed in table 1. For AA-stacking (figure 1(a)), the second layer is exactly above the first one. But for the AB-stacking (figure 1(b)), the upper layer is shifted along the xy plane relative to the lower layer. For AC-stacking (figure 1(c)), the upper layer is not only moved along the xy plane but is also rotated relative to the lower layer. The interlayer distances in the bilayer structures are $3.214 \text{ \AA}$, $3.248 \text{ \AA}$ and $3.262 \text{ \AA}$ for AA-, AB- and AC-stackings, respectively (figure S1, available online at stacks.iop.org/JPhysD/52/385103/mmedia). The AA-stacking BBP possesses the shortest distance between layers and the smallest band gap. The lattice constants and binding energy of AA- and AB-stacking BBP are similar.

Electronic band structures are plotted in figure 2 for monolayer BlueP as well as for AA-, AB-, and AC-stacking BBP.

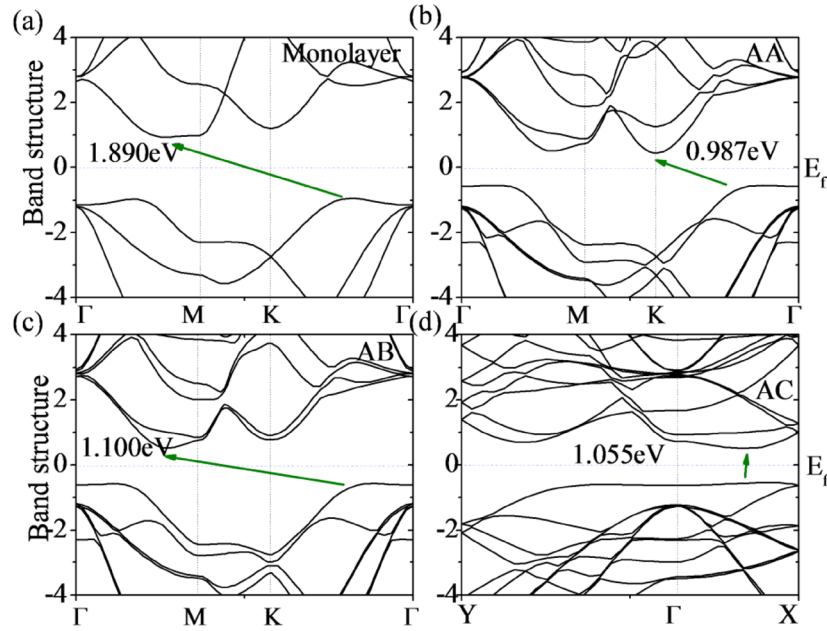


Figure 2. The band structure of the monolayer and BBP.

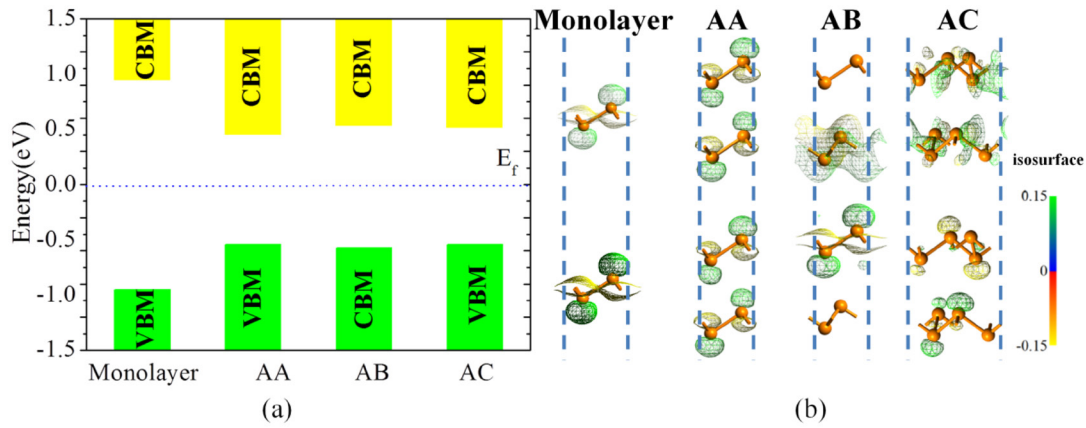


Figure 3. Energy levels of conduction band offset (CBM), and valence band offset (VBM) of monolayer and bilayer BlueP. (c) Isosurface plot of orbitals from molecular projected self-consistent Hamiltonian (MPSH) of BBP. The crystalline orbitals of the monolayer and bilayer BlueP calculated for the Γ point corresponding to CBM and VBM. The isosurface is $0.15 \text{ \AA}^{-3/2}$.

Each energy band in the monolayer is split into two bands due to interactions between the two adjacent layers. Otherwise, the overall shape of the valence band maximum (VBM) and the conduction band minimum (CBM) are hardly changed except for AC-stacking BBP compared with monolayer BlueP, but the band gap is decreased to about 1 eV. For bilayer structures, the p orbital enters the original electronic gap region.

The calculated binding energies (E_b) for three types of stacking constructions are summarized in table 1. From the results, the AB-stacking is the most favorable stacking structure. For pure BBP, our results are consistent with those of Ghosh *et al* [19]. In addition, the lattice constant in the plane is stretched by intercalation, but due to the strong covalent bond of intralayer P–P bonds, the length of P–P bonds is almost unchanged. As illustrated in figure 3, we calculate the band gap between the conduction and valence bands as summarized in table 1. The electronic band structure of monolayer BlueP is qualitatively similar for both GGA and LDA based

calculations. They give 1.89 eV (GGA) and 1.73 eV (LDA) band gaps. In particular, the value from GGA is in accordance with those reported in [7]. Similarly, BBP is also an indirect energy-gap semiconductor, and the energy gap is about 1 eV for all three stacking sequences (AA, AB, and AC) (see figures 2(b)–(d)). The value of the band gap of BBP is stacking sequence dependent: 0.987 eV for AA-, 1.100 eV for AB-, and 1.055 eV (GGA) for AC-stackings, which is 41%–47% less than that of monolayer BlueP.

We also plot the alignment of CBM and VBM with respect to the Fermi level, as shown in figure 3(a), while figure 3(b) shows the corresponding crystalline orbitals. It can be seen that VBMs are similar in the three superimposed modes, while the CBMs of AA superimposed BBP are significantly lower than that of AB and AC. Also, the overlap of interlayer crystalline orbitals of CBM is absent in stacked bilayers. The electronic orbital state is almost located on one layer for AB-stacking BBP, suggesting the weaker interlayer interactions in AB-stacking BBP.

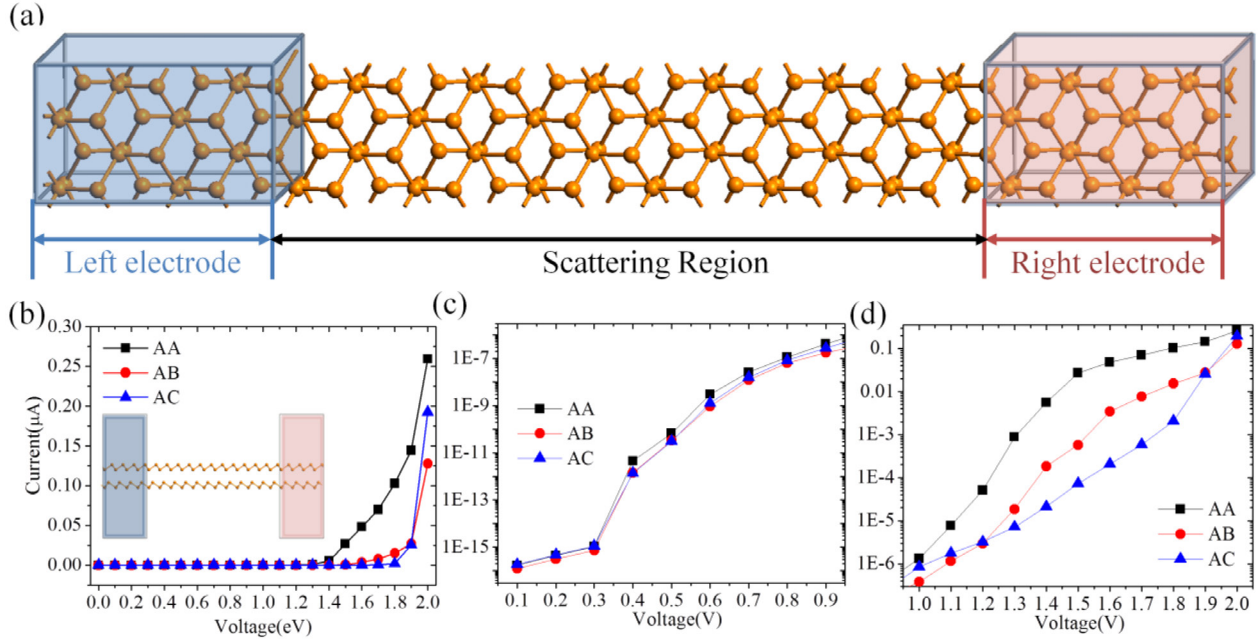


Figure 4. (a) Structures of AB stacking bilayer BlueP schematic device model of the two probe system. The blue frame indicates the semi-infinite left leads while the pink one indicates the semi-infinite right leads. (b) The current–voltage curve of AA-, AB- and AC-stacking bilayer BlueP schematic device. (c) To observe the variation of current at low bias, we also plot the I – V characteristics of AA-, AB- and AC-stacking bilayer BlueP in semi-log scales of the bias voltage range of [0–1.0 V]. (d) I – V characteristics in semi-log scales of the bias voltage range of [1.0–2.0 V].

We study transport structures of the BBP with atomic positions all determined by the geometry optimization without fixing any phosphorus atoms. It is composed of a scattering region and left, right electrode contacts as shown in figure 4(a). We have optimized the sandwich structures. The electron-dependent transport characteristics of these systems are studied by using DFT and NEGF. In our electric transport calculations, the valence orbitals are expanded in a double- ζ plus polarization (DZP) basis set for all atoms. For the electrodes in Brillouin zone (BZ), the $1 \times 1 \times 100$ k -points mesh is applied and the cut-off energy is set to 150 Ry for the real space grid.

The current is calculated as

$$J_{\sigma} = \int_{\mu_R}^{\mu_L} T_{\sigma}(E, V) [f(E - \mu_L) - f(E - \mu_R)] dE \quad (1)$$

where μ_L and μ_R are the chemical potentials of the left and the right electrodes, while f is the Fermi–Dirac distribution.

We simulate a device structure as shown in figure 4(a), where the scattering region is the centre region and the left and right electrodes are semi-infinite BBP. The electronic transport direction is from left to right and, along the z direction, the left and right electrodes are all set to room temperature—300 K. The calculated I – V curves of the AA-, AB- and AC-stackings BBP are as shown in figure 4(b); to observe the variation of current at low bias, we also plot the I – V characteristics of AA-, AB- and AC-stacking bilayer BlueP in semi-log scales in the set of figures 4(c) and (d), respectively. In the bias voltage range of [0–1.0 V], the currents of the three devices are similar (see figure 4(c)). The current of the AA-stacking device grows quickly while the bias voltage is increased further. The current of the AC-stacking device grows slowest.

When the bias voltage is larger than 1.9 V, the current of the AC stacking device suddenly increases and exceeds the AB one.

In order to explain the transport behaviors in figure 4, we have also studied the transmission spectra dependent bias voltage and energy level which are summarized in figure 6. For equation (1), the current is defined by the integral of the transmission coefficient and Fermi energies within the bias window. We show the transmission spectra of the AA-stacking (figure 5(a)) that when the bias voltage is in the range from 0 to 1.3 V, in the bias window, the transmission spectra of the transmission channel is suppressed. So, the current in region I is very low. When the bias increases, the peaks of the transmission spectra move into the bias window and the current of AA-stacking increases rapidly as shown in region II. For the AB-stacking device, the current is in the off state in the bias range from 0 to 1.5 V. When the bias voltage increases further, the current is turned on and increases rapidly. The current-off state of the AC-stacking device is attained at a bias voltage from 0 V to 1.7 V. We see a more rapid increase of the current in the AC stacking device for bias beyond 1.7 V.

Figure 6 shows the transmission coefficients at four particular bias voltages (0 V, 1.0 V, 1.6 V and 2.0 V) for the BBP devices. For AA-stacking (see figure 6), only when the bias is 1.6 V in the bias window is there a transmission peak, leading to the increase of the current. Then, when the bias voltage takes the value 2.0 V (see figure 6), the bias window is enlarged by the increase of the bias voltage, due to the total integral area and the corresponding transmission coefficient becoming larger, which stands for the fact that the corresponding orbital is opened. In the issue, the current in the AA-stacking device is increased. As shown in figure 6, for AC-stacking, there is no

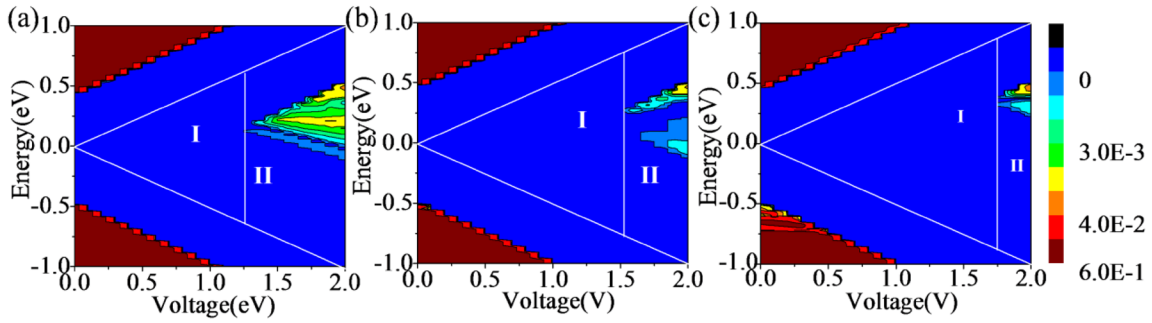


Figure 5. The transmission coefficients as a function of electron energy E and bias voltage for (a) AA-, (b) AB- and (c) AC-stacking bilayer BlueP. Note that region I and region II are bias windows.

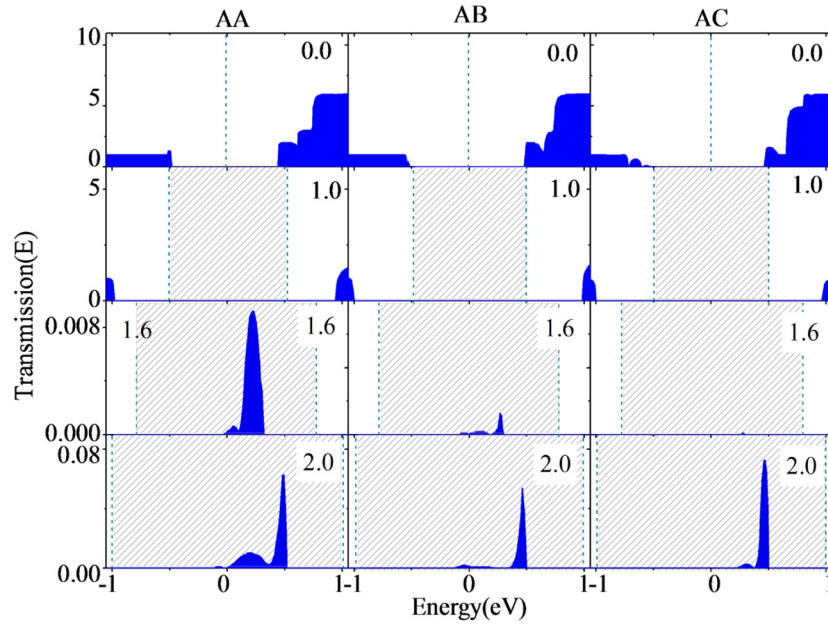


Figure 6. The transmission spectra of bilayer BlueP at bias voltage 0.0 V, 1.0 V, 1.6 V and 2.0 V.

transmission peak in the bias window at the 1.6 V, while some small transmission peaks move into the bias window for the AB one, so the current for AB-stacking exceeds that for the AC one. When the bias voltage reaches 2.0 V, the transmission coefficient for AC-stacking is higher than the AB one, and the current of AC exceeds that of the AB one. When the bias voltage is in the range from 1.0 to 2.0 V, there is an interesting competitive relationship between the currents of AB- and AC-stacking devices.

As mentioned above, the behaviors of the currents can be explained based on the transmission spectra, and the origin of the transmission spectra should be correlated with the electron transmission from left to right electrode and will be discussed later. It has been reported that the transmission channel is either open when the symmetries of the electronic subbands of the left and right electrodes match, or it is closed while the energy levels of the left and right electrodes are mismatched at the same energy region [20, 21].

To understand the transmission spectra, we take two particular bias voltages for 1.6 V and 2.0 V and three bilayer configurations (AA, AB, and AC) for further study. Figure 7 shows the transmission coefficient (middle panel) and band

structures of the left electrode (left panel) and right electrode (right panel) at different bias voltages. When the bias is applicable to the left and right electrodes, a positive bias shifts the energy bands up; meanwhile, a negative bias shifts the energy bands down. This leads to a change in the symmetrical matching of the left and right electrodes. Due to the buckling and bilayer structures of BlueP, we cannot find any simple π or σ bonds or antibonds close to the Fermi level. The grey shadow represents the efficacious transmission channel within the bias window (light-grey), which has a significant impact on the electronic transmission, and the grey region denotes that the channel is open for electron flow. Thereupon, from figure 7, the results show that the voltage–current curve and transmission spectrum of the electrodes are in good agreement with the symmetry of the electrode’ band structures.

For the AA-stacking structure, at bias voltage of 1.6 V (as shown in figure 7(a)), the xz plane mirror symmetry operation relates the lowest unoccupied molecular orbital (LUMO)-1 of the left electrode (LUMO-1(L)) and the highest occupied molecular orbital (HOMO) of the right electrode (HOMO(R)). Due to the symmetrical matching of the left and right electrodes, the channel opens within the energy range of [0.05–0.35 eV]

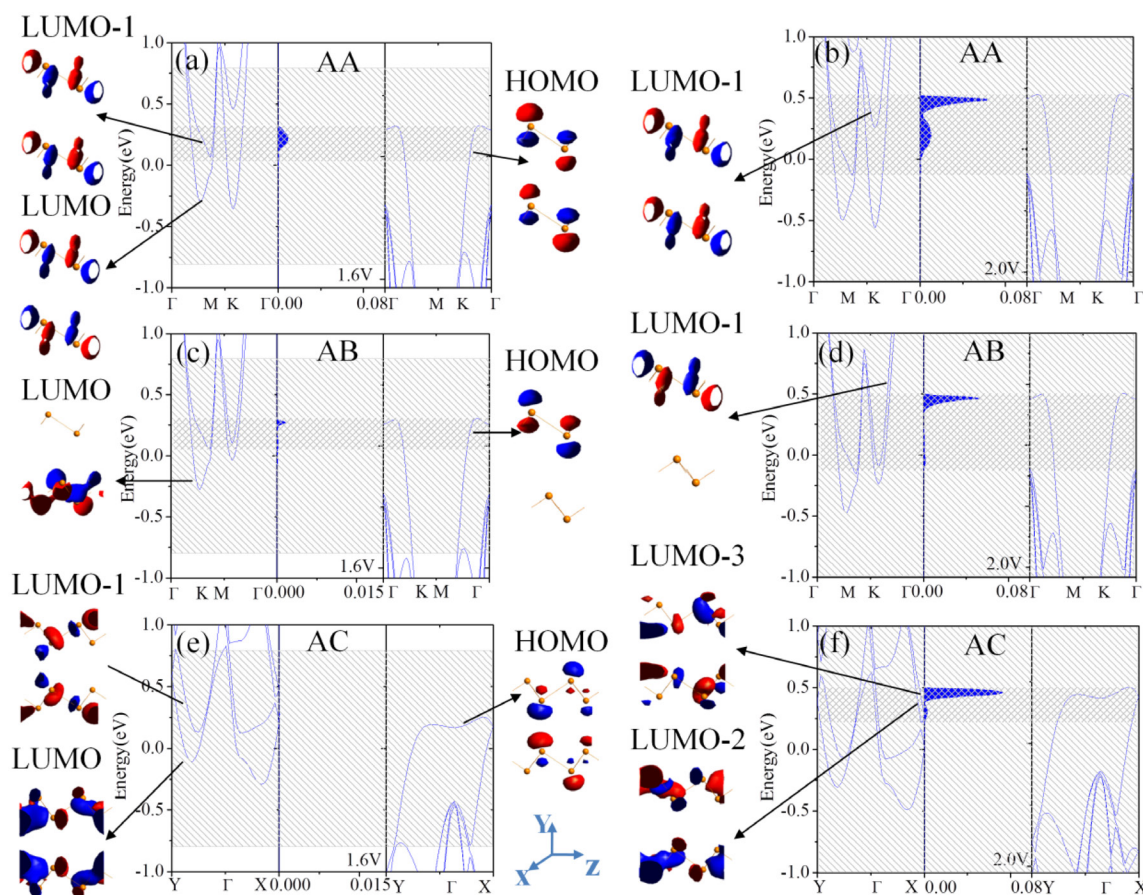


Figure 7. The band structures of left electrode (left panel) and right electrode (right panel), along with the transmission spectra (middle panel), for a bias of 1.6 V and 2.0 V, separately. The light-grey shadow represents the bias window and the dark-grey shadow represents the efficacious transmission channel. The corresponding electronic orbitals are also shown.

in the dark gray region, while the channel closes in the rest of the bias window due to the symmetry mismatch. The symmetry matched region of LUMO-1(L) and HOMO(R) is large, leading to the increase of the current in figure 7(b).

In the AB-stacking configuration, only the LUMO-1(L) and HOMO(R) have the same symmetry under the Y axis. Similar to AA-stacking, the down-shifted LUMO-1(L) matches the up-shifted HOMO(R) and a wider region in the transmission spectra contributes to the current by increasing the bias voltage.

In the AC-stacking configuration (as shown in the figure 7(e)), there is no transmission spectrum in the bias window because of the different symmetries of LUMO-1(L) and LUMO (L) compared with the HOMO(R). However, when the bias is 2.0 V, the LUMO-2(L) and LUMO-3(L) which are of the same symmetry with the HOMO(R) enters the bias window, causing higher transmission than the AB one. Therefore, the current of AC-stacking exceeds that of the AB one at large bias voltage.

Summary

In summary, we have systematically investigated the structural and electronic characters of AA-, AB- and AC-stacking BBP by means of DFT. We have also demonstrated the nonlinear

current behavior of different BBP by first principles calculations based on DFT combined with the NEGF technique. We construct device structures by sandwiching the BBP with the 2D stacking BBP electrodes and different stacking show different conductivity. The AA-stacking BBP possesses the best conductivity due to the smallest band gap. Because of the symmetry of the electronic subbands, the current of either AC- or AB-stacking BBP can be larger depending on the bias voltage, showing the interesting competition behaviors. The BBP with interesting electronic and transport properties are expected to have potential applications in nanoelectronics.

Acknowledgments

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